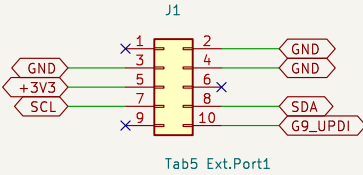
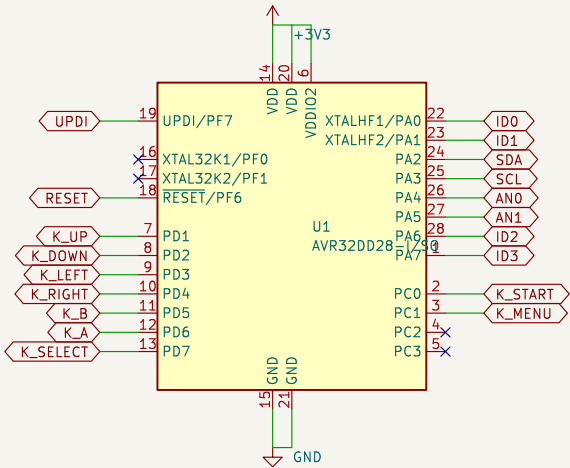


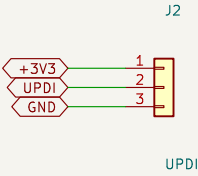
MCU – AVR32DD28 (SOIC–28) running firmware_avrdd (I2C client 0x6D, RetroPad Ext. Port 1) – verify pin 1 orientation before fab

Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware_avrdd/pinmap.h

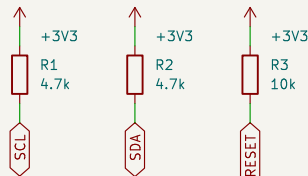
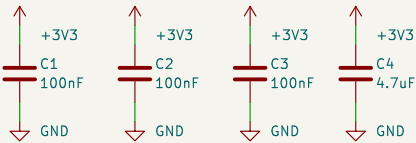
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



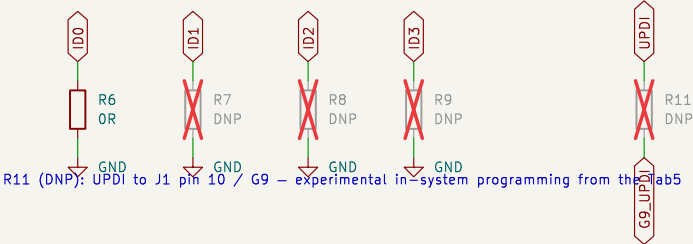
UPDI programming (SerialUPDI / MPLAB Snap)



Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up

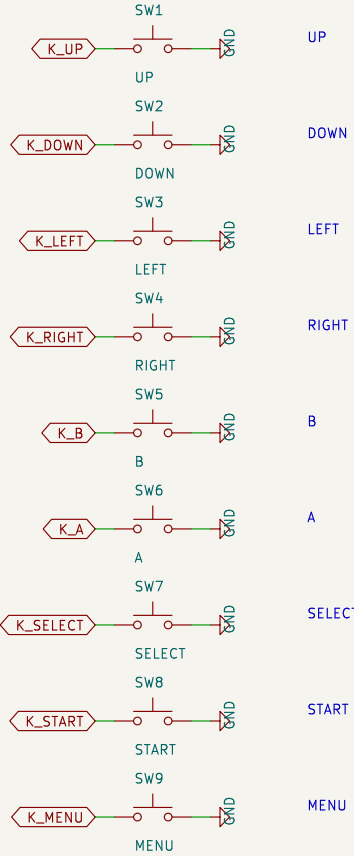
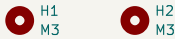


Console ID straps: OR to GND = bit set. NES (AVR DD) = ID 1 (fit R6: R7, R8, R9 DNP)



R11 (DNP): UPDI to J1 pin 10 / G9 – experimental in–system programming from the Tab5

Mounting holes (M3, 96 mm apart)



Tab5 keyboard–port board: STM32F030 + RetroPad firmware, I2C 0x6D
Generated by kicad_gen/gen_sch.py from gen_pcb.py and layout.py

RetroESP32–P4

Sheet: /
File: retropad_nes_dd.kicad_sch

Title: RetroPad NES (AVR DD) controller (console ID 1)

Size: A3
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Date: 2026–10–05

Rev: 0.1

Id: 1/1